

Jiahui Qu

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EDUCATION

London School of Economics and Political Science

MSc Statistics (Financial Statistics)

Selected marks: Artificial Intelligence (87/100); Managing and Visualising Data (84/100); Reinforcement Learning (83/100)

London, UK

Sep. 2025 – Mar. 2027

University of Edinburgh

BSc (Hons) Mathematics and Statistics

Exchange: City University of Hong Kong, *Computing Mathematics* (2023–2024)

GRE: 338/340 (V168, Q170), Analytical Writing 4.0.

Edinburgh, UK

Sep. 2021 – Jun. 2025

RESEARCH INTERESTS

Human–AI Interaction; AI-supported Learning; Learning Analytics; Language Models; Statistical Machine Learning.

PUBLICATIONS & MANUSCRIPTS

- [1] **Jiahui Qu**, Yifang Qin, Boyang Zheng, Ziyi Zhou. “Calibrate Once, Choose the Beam: A Predictive Regime Test for Same-LM Search Guidance and Pruning.” *ICML 2026 LM4Plan Workshop (accepted)*.
- [2] **Jiahui Qu**, Yifang Qin, Boyang Zheng, Ziyi Zhou. “Calibrate Once, Choose the Beam: A Pre-Deployment Compute-Allocation Rule for Same-LM Multimodal Search.” *ICML 2026 SCALE Workshop (accepted)*.
- [3] **Jiahui Qu**, Boyang Zheng, Yifang Qin, Yudi Cao, Jiayu Fang. “Screen, Confirm, Then Search: Risk-Controlled Pre-Deployment Qualification for LM-Guided Search.” *TMLR (under review)*.
- [4] **Jiahui Qu**, Shizhao Xiong, Jiayu Fang, et al. “When Better Scores Select Worse Boards: Replacement-Level Audits for Teamfight Tactics Search.” *Under review*.
- [5] **Jiahui Qu**, Shizhao Xiong, Yifang Qin, et al. “When Evidence Drifts: Evidence-State Routing and Guarded Revision Release for Long-Form Factual Generation.” *Under review*.

RESEARCH EXPERIENCE

Lasso-to-Learn: Adaptive AI Intervention for Learning

Dec. 2025 – Present

Independent Research Project | YNotes

- Model learner state from reading and note-taking context and estimate the counterfactual effects of wait, probe, and hint actions, with the goal of learning minimally sufficient intervention policies.
- Develop source-anchored interventions that provide contextual feedback and follow-up practice while preserving the learner’s ongoing reading and note-taking workflow.
- Design counterfactual evaluations and human-subject studies to compare alternative intervention policies and identify when AI assistance is beneficial or unnecessary.

A Study on Hoeffding’s D-Test for Independence

May. 2024 – Aug. 2024

Funded Research Student | University of Edinburgh | Supervisor: William Scott

- Studied Hoeffding’s D-statistic for detecting non-monotonic dependence and investigated its theoretical relationship with Even-Zohar’s representation.
- Applied dependence testing and bootstrap inference to financial-stability data from 13 African countries, with statistical and machine-learning baselines for systemic-crisis prediction.

Moments of Random Matrices and Combinatorics

Apr. 2023 – Sep. 2023

Funded Research Student | University of Edinburgh | Supervisor: Theo Assiotis

- Studied asymptotic questions related to the Keating–Wong conjecture for moments of random matrices and log-correlated fields.
- Derived and computationally verified expressions involving Gamma-function sums, using symbolic and numerical experiments to examine large-N asymptotics.

SOFTWARE & SYSTEMS

YNotes — Founder & Developer

2025 – Present

AI-native iPad notebook for source-grounded learning | App Store | ynotesapp.com

- Designed and shipped a handwriting-first learning workflow integrating PDF reading, Apple Pencil annotation, lasso-selected AI tutoring, in-place explanations, and generated practice.
- Built the end-to-end iPad and cloud system in SwiftUI/PencilKit, supporting contextual AI interactions, persistent notes and annotations, and production deployment.
- Released YNotes on the Apple App Store and use it as the experimental platform for studying adaptive AI intervention in Lasso-to-Learn.

HONOURS & AWARDS

- **LSE Annual Scholarship**, London School of Economics and Political Science 2025
- **Undergraduate Vacation Scholarship**, University of Edinburgh 2023 & 2024
- **Kaggle Silver Medal — GoDaddy Microbusiness Density Forecasting**, Top 3% 2023
- **Second Prize, 18th Citi Financial Innovation Application Competition**, Top 1% 2023
- **Edinburgh Leadership Award**, University of Edinburgh 2023

LEADERSHIP & MENTORING

Peer Support Leader, University of Edinburgh 2022 - 2023

- Led after-class problem-solving workshops on core mathematics coursework and competition-style problems, helping junior students develop independent study and problem-solving strategies.
- Mentored junior students and organised peer-learning and social activities, helping them build academic confidence, peer connections, and a smoother transition to university study.

Department Lead, Lambda — AI & Computing Society, University of Edinburgh 2021 – 2022

- Led Lambda's AI/computing department, coordinating its student team and technical programme.
- Organised an AI/computing hackathon for 50+ participants, coordinating team formation, event delivery, and technical support.

PROFESSIONAL EXPERIENCE

Guoyuan Securities Jun. 2024 – Sep. 2024

Quantitative Equity Investment Intern | Wuhan, China

- Developed and backtested ETF-rotation strategies using momentum, trend, and cross-asset correlation signals for dynamic portfolio allocation.
- Automated position adjustment using 5-day rebalancing and 20-day trend evaluation, and assessed strategies using return, Sharpe ratio, and drawdown metrics.

Guotai Junan Securities Jun. 2022 – Sep. 2022

Quantitative Researcher Intern | Shanghai, China

- Built portfolio and factor analytics in Python and R, including return, volatility, maximum-drawdown, and P&L attribution analysis.
- Constructed market-anomaly factors from Wind data and analysed A-share electronics-sector valuation and market-cap trends.

PATENT

Data Classification Framework for Financial Economics 2023
Patent No. ZL 2023 3 0301344.6

TECHNICAL SKILLS

Programming: Python, R, MATLAB, Swift/SwiftUI, SQL.

Machine Learning & Statistics: PyTorch, Hugging Face, scikit-learn, LightGBM, statistical modelling, bootstrap inference, experimental design.

Systems & Tools: PencilKit, Git, LaTeX, Cloudflare Workers.

Languages: Mandarin Chinese (native), English (fluent).